



HGLTR: Hierarchical Knowledge Injection for Calibrating Pre-trained Models in Long-Tail Recognition

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Background

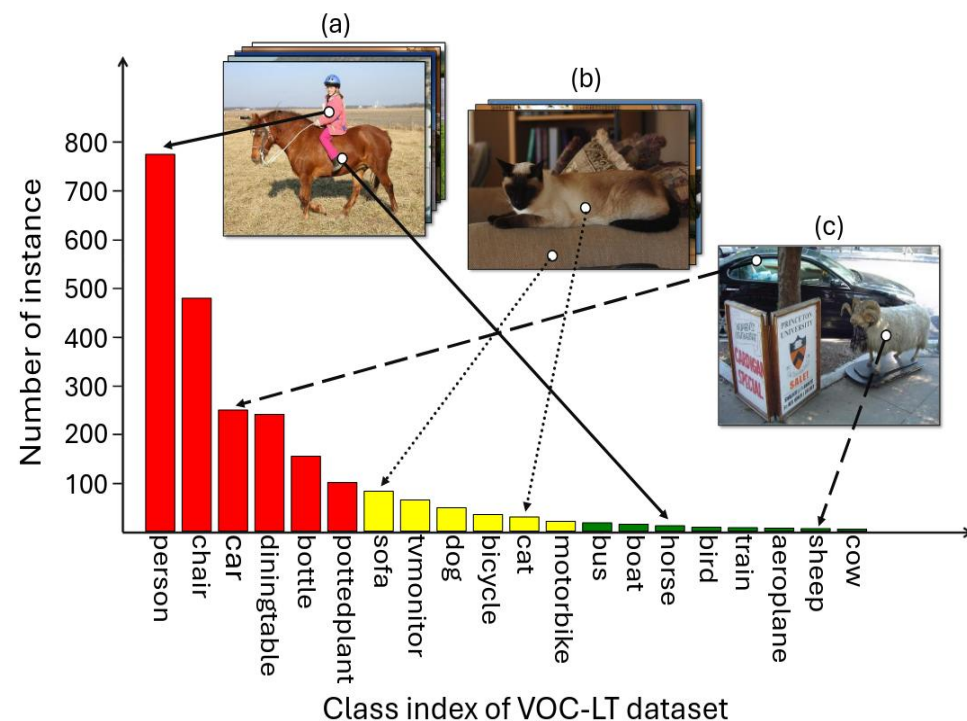


Figure 1: An illustrative example and the challenges of the LT-MLC problem on the VOC-LT dataset(Wu et al. 2020).

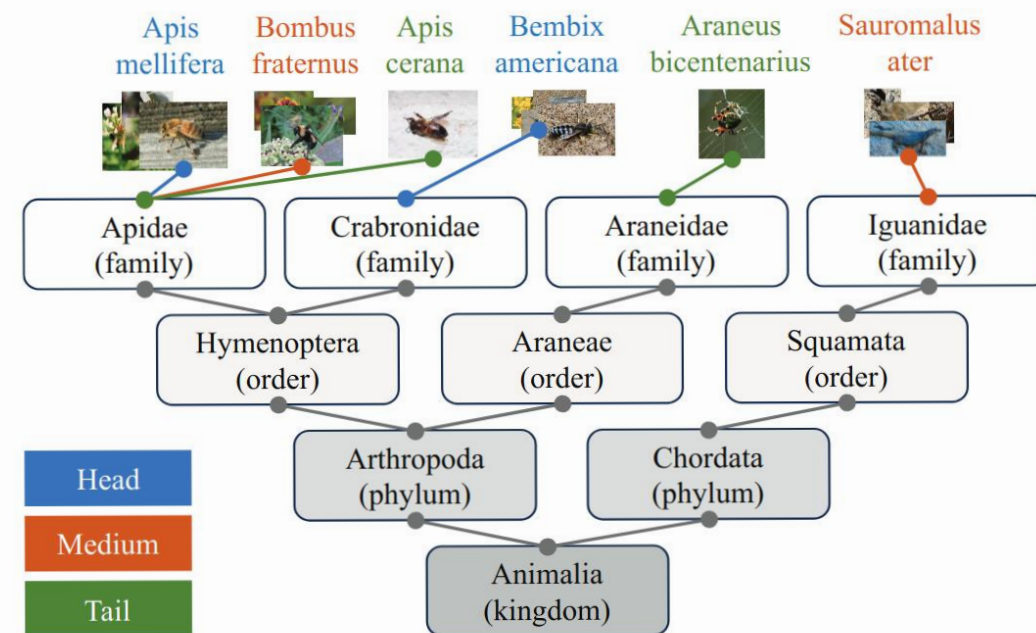
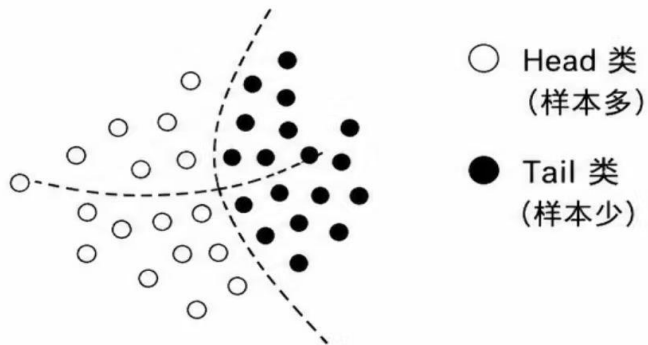


Figure 1: Using the hierarchical structure from the iNaturalist 2018, we illustrate the hierarchical semantic relationships among different distribution categories.

1. 传统长尾学习方法 (From-scratch Models)

主要做法

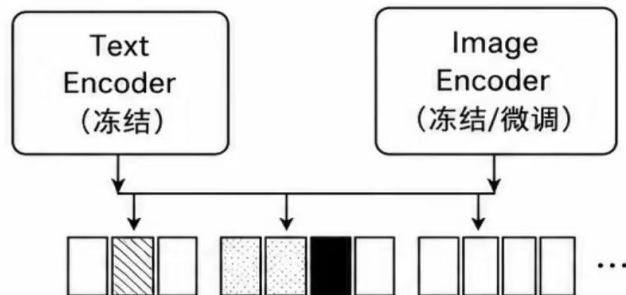
- 重采样 / 类别重加权
- 损失函数设计 (focal loss, LDAM, LA 等)
- 多专家 / 元学习等



2. 基于预训练模型的长尾学习 (如 CLIP/VLM 方法)

主要做法

- 使用 CLIP 等预训练模型作为 backbone
- 轻量微调: Adapter / Prompt / Decoder
- 损失或 logits 调整:
Logit Adjustment, Balanced Loss,
Cross-modal 融合等



主要发现

- 预训练数据 (互联网图文) 本身长尾
- 预训练模型存在 label bias / data bias (Zhu et al., 2023 等)
- 在长尾微调时, 容易过预测 head 类, 欠拟合 tail 类

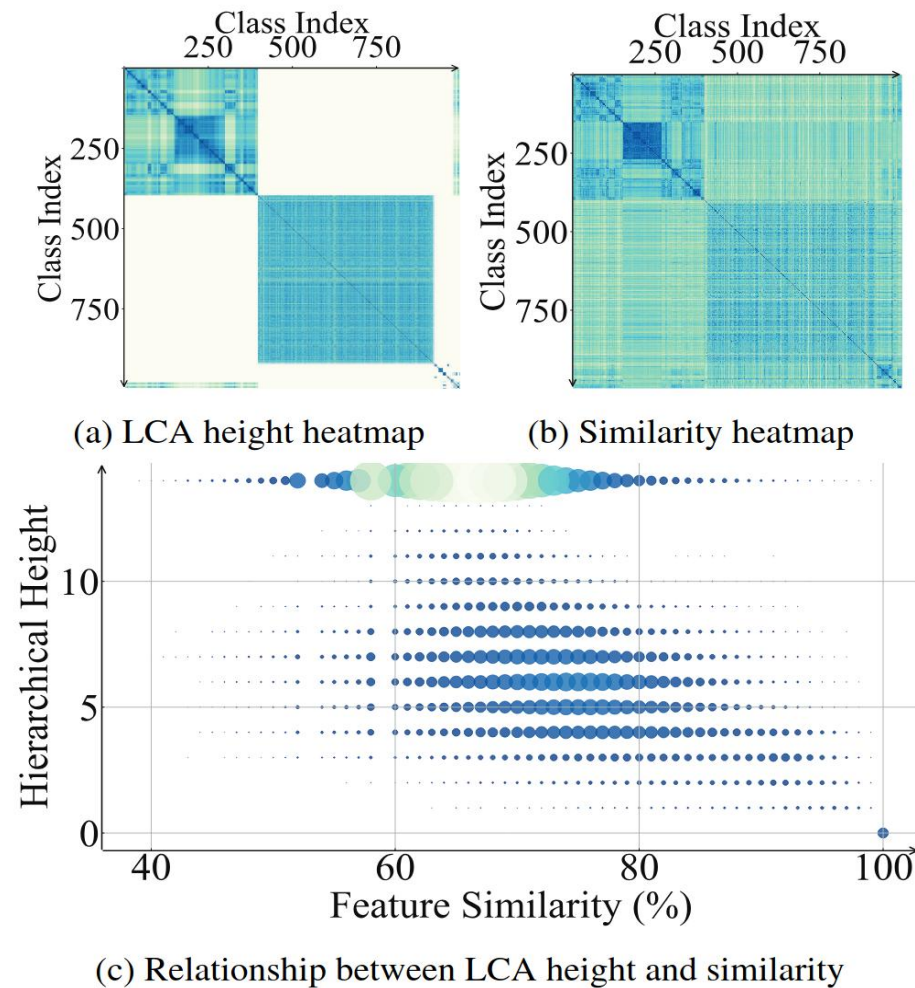


Figure 3: Hierarchical structure and feature space analysis on ImageNet. (a) LCA height heatmap: darker colors indicate smaller semantic distance. (b) Feature similarity heatmap: darker colors indicate higher cosine similarity between class prototypes. (c) Scatter plot showing a negative correlation between LCA height (y-axis) and feature similarity (x-axis); point size denotes sample pair counts.

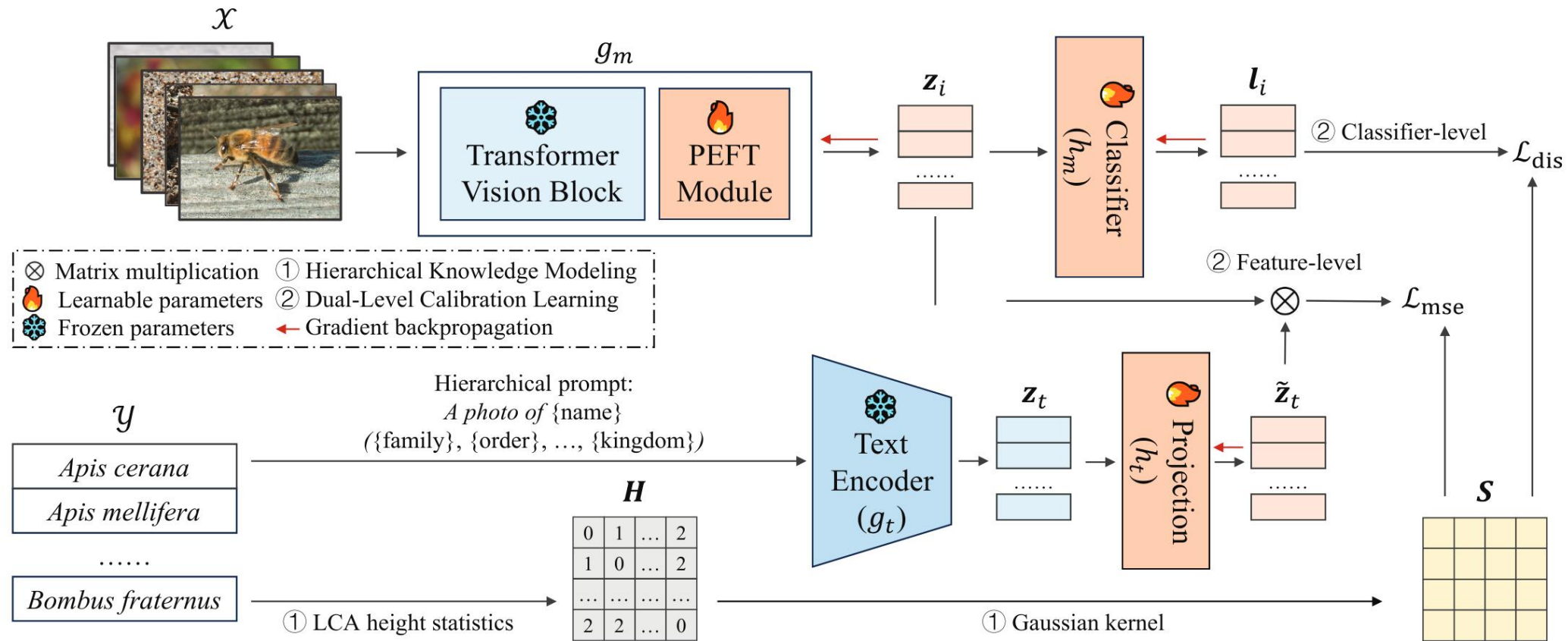


Figure 2: HGLTR's two-stage framework: ① Hierarchical Knowledge Modeling constructs the semantic blueprint by computing LCA-based distance matrix H from class hierarchy and converting it to hierarchical similarity matrix S ; ② Dual-Level Calibration Learning fine-tunes the model: at the feature level, image features z_i from frozen CLIP backbone with PEFT modules are aligned with S using $\mathcal{L}_{mse}$; at the classifier level, predictions are regularized with soft labels from S via $\mathcal{L}_{dis}$.

Hierarchical Knowledge Modeling

$$S_{k_1, k_2} = \frac{\exp(-H_{k_1, k_2} \cdot \tau_s)}{\sum_{k=1}^K \exp(-H_{k_1, k} \cdot \tau_s)}, \quad (1)$$

$$S_{k_1, :} = \text{softmax}(-\tau_s H_{k_1, :})$$

	dog	cat	car	bus
dog	0	1	2	2
cat	1	0	2	2
car	2	2	0	1
bus	2	2	1	0

```
s_dog = [0.610, 0.225, 0.083, 0.083]
```

```
raw score = exp(-H)
```

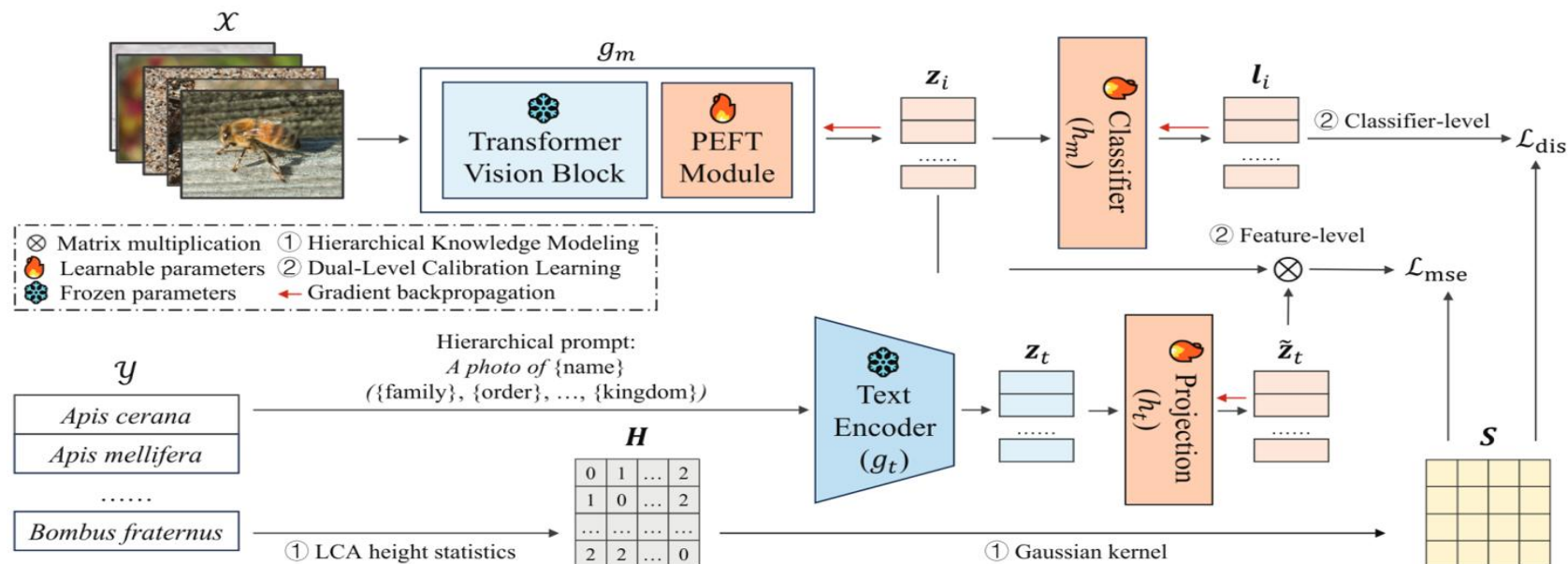
```
dog: exp(0) = 1
```

```
cat: exp(-1) = 0.3679
```

```
car: exp(-2) = 0.1353
```

```
bus: exp(-2) = 0.1353
```


Feature-level



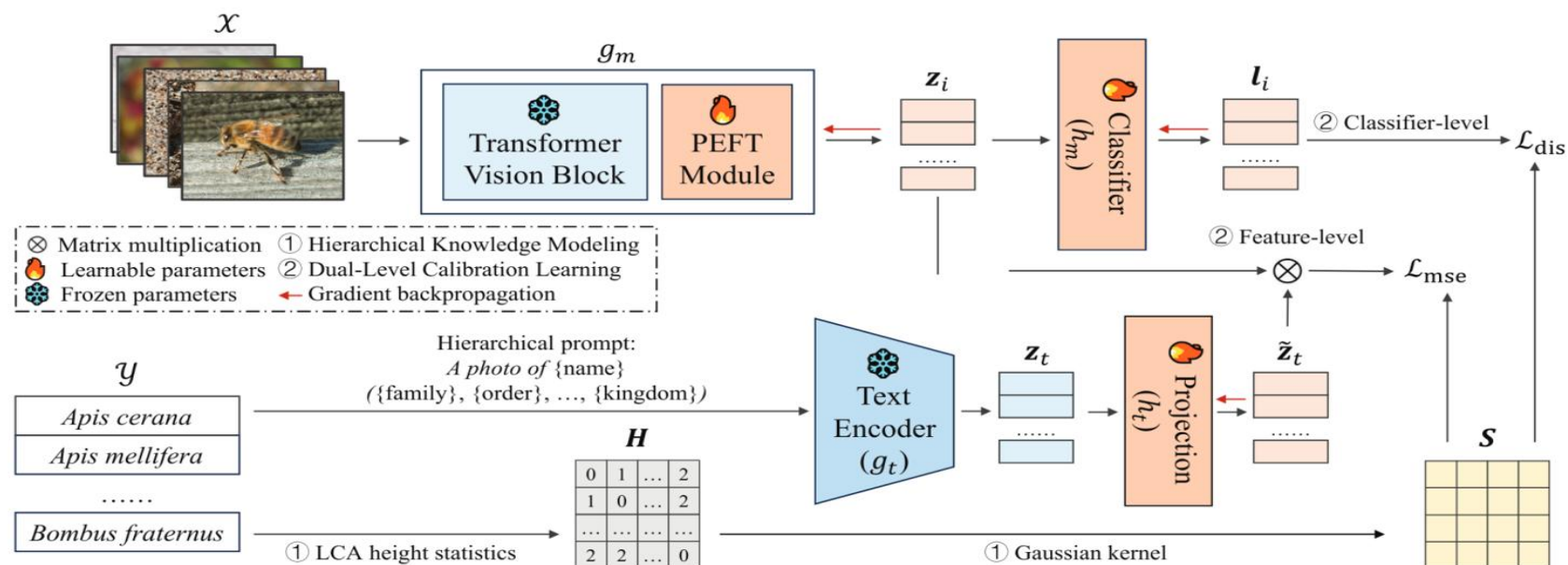
$$\tilde{z}_t = h_t(z_t) = z_t W_t^\top + b_t. \quad (2)$$

$$\mathcal{L}_{mse} = \frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} (\text{sim}(z_i, \tilde{z}_t) - S_{y_i})^2, \quad (3)$$

a photo of a Apis cerana (Apidae, Hymenoptera, Insecta, Animalia)

a photo of a {class name} ({family}, {order}, ..., {kingdom})

Classifier-level



$$\mathcal{L}_{dis} = \frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} \text{KL}(p_i \| s_{y_i}). \quad (4)$$

$$p_i = \text{softmax}(l_i / \tau_l)$$

$$L_{dis} = \frac{1}{\tilde{N}} \sum_{i \in \mathcal{I}_{nt}} \sum_{k=1}^K p_{i,k} \log \frac{p_{i,k}}{s_{y_i,k}}$$

$$s_{dog} = [0.610, 0.225, 0.083, 0.083]$$

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(l_{i,y_i} + \log q_{y_i})}{\sum_{k=1}^K \exp(l_{i,k} + \log q_k)}, \quad (5)$$

head 类: $q = 0.9$, $\log q = -0.105$

tail 类: $q = 0.1$, $\log q = -2.303$

head adjusted logit = $2.0 - 0.105 = 1.895$

tail adjusted logit = $2.0 - 2.303 = -0.303$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \alpha \mathcal{L}_{\text{mse}} + \beta \mathcal{L}_{\text{dis}}, \quad (6)$$

Methods	Backbone	Learnable Params.	Epochs	Overall	Head	Medium	Tail
Training from scratch							
MiSLAS (Zhong et al. 2021)	ResNet-50	23.51M	180+10	52.7	62.9	50.7	34.3
LA (Menon et al. 2021)	ResNet-50	23.51M	90	51.1	-	-	-
DisAlign (Zhang et al. 2021)	ResNet-50	23.51M	90	52.9	61.3	52.2	31.4
NCL (Li et al. 2022b)	ResNet-50	23.51M	400	57.4	-	-	-
ProCo (Du et al. 2024)	ResNet-50	23.51M	180	57.8	68.2	55.1	38.1
CBTS (Zhao et al. 2025)	ResNet-50	23.51M	90	59.8	70.0	57.3	39.1
LiVT (Xu et al. 2023)	ViT-B/16	85.80M	100	60.9	73.6	56.4	41.0
Fine-tuning foundation model							
BALLAD (Ma et al. 2021)	ViT-B/16	149.62M	50+10	75.7	79.1	74.5	69.8
Decoder (Wang et al. 2024)	ViT-B/16	21.26M	~18	73.2	-	-	-
HGLTR(Ours)	ViT-B/16	1.61M	10	77.4	80.4	76.5	72.1
Fine-tuning with extra data							
VL-LTR (Tian et al. 2022)	ViT-B/16	149.62M	100	77.2	84.5	74.6	59.3
GML (Suh and Seo 2023)	ViT-B/16	149.62M	100	78.0	-	-	-

Table 2: Comparison performance (Top-1 Accuracy%) on ImageNet-LT.

Methods	Backbone	Learnable Params.	Epochs	Overall	Head	Medium	Tail
Training from scratch (with an ImageNet-1K pre-trained backbone)							
OLTR (Liu et al. 2022)	ResNet-152	58.14M	30	35.9	44.7	37.0	25.3
MiSLAS (Zhong et al. 2021)	ResNet-152	58.14M	90+10	40.4	39.6	43.3	36.1
DisAlign (Zhang et al. 2021)	ResNet-152	58.14M	30	39.3	40.4	42.4	30.1
ALA (Zhao et al. 2022)	ResNet-152	58.14M	30	40.1	43.9	40.1	32.9
LiVT (Xu et al. 2023)	ViT-B/16	85.80M	100	40.8	48.1	40.6	27.5
Fine-tuning foundation model							
BALLAD (Ma et al. 2021)	ViT-B/16	149.62M	50+10	49.5	49.3	50.2	48.4
Decoder (Wang et al. 2024)	ViT-B/16	21.26M	~34	46.8	-	-	-
LPT (Dong et al. 2023)	ViT-B/16	1.01M	40+40	50.1	49.3	52.3	46.9
GNM-PT (Li et al. 2024)	ViT-B/16	1.01M	100	50.1	46.6	53.3	49.4
HGLTR(Ours)	ViT-B/16	1.61M	10	51.4	51.1	51.6	51.2
Fine-tuning with extra data							
VL-LTR (Tian et al. 2022)	ViT-B/16	149.62M	100	50.1	54.2	48.5	42.0
RAC (Long et al. 2022)	ViT-B/16	85.80M	30	47.2	48.7	48.3	41.8

Table 3: Comparison performance (Top-1 Accuracy%) on Places-LT.

Thanks